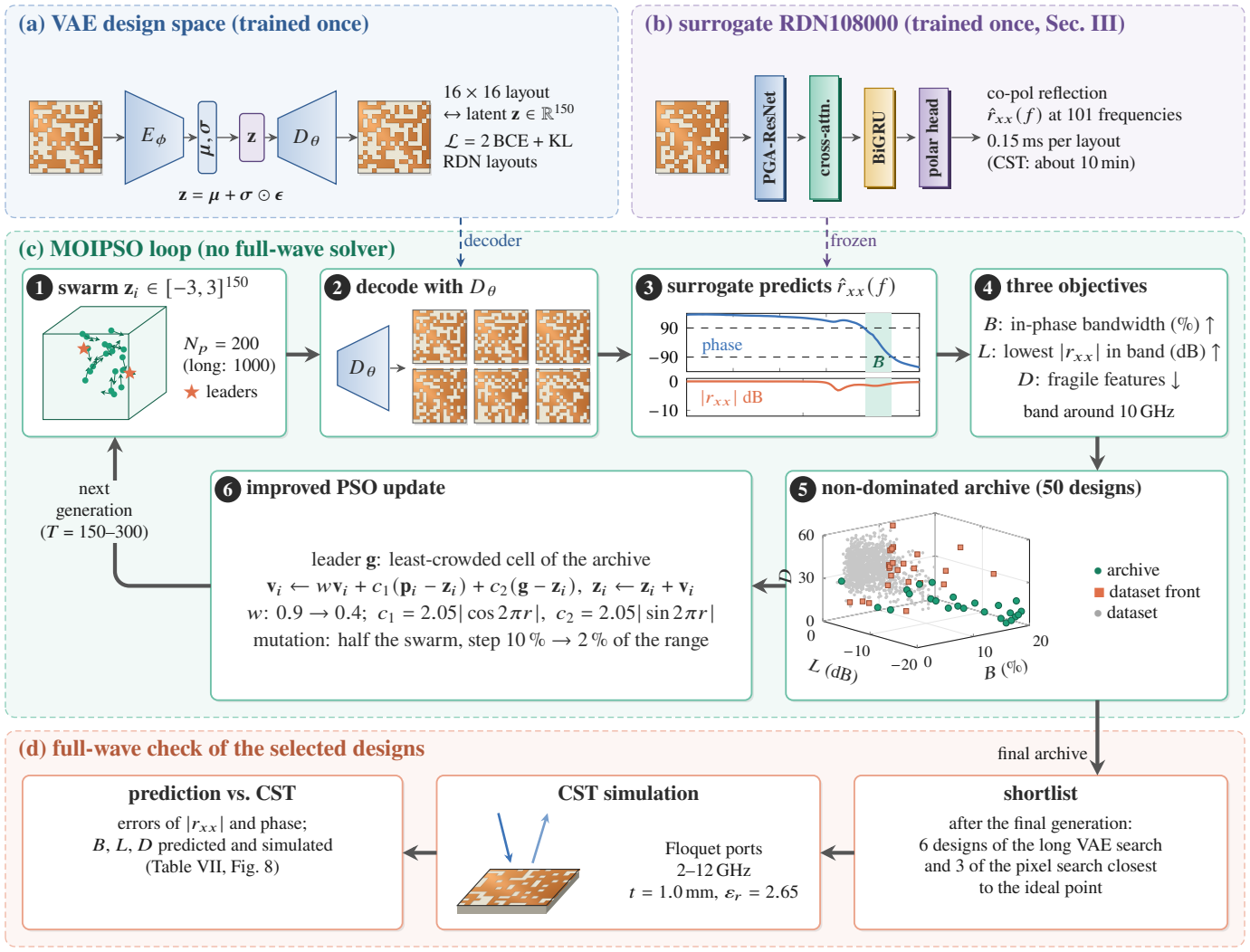

(a) VAE design space (trained once)
E_ϕ
μ, σ
z
D_θ
$\mathbf{z} = \mu + \sigma \odot \epsilon$
16 × 16 layout
↔ latent $\mathbf{z} \in \mathbb{R}^{150}$
$\mathcal{L}$ = 2 BCE + KL
RDN layouts
(b) surrogate RDN108000 (trained once, Sec. III)
PGA-ResNet
cross-attn.
BiGRU
polar head
co-pol reflection
$\hat{r}_{xx}(f)$ at 101 frequencies
0.15 ms per layout
(CST: about 10 min)
decoder
frozen
(c) MOIPSO loop (no full-wave solver)
1 swarm $\mathbf{z}_i \in [-3, 3]^{150}$
N_p = 200
(long: 1000)
leaders
2 decode with D_θ
D_θ
3 surrogate predicts $\hat{r}_{xx}(f)$
90
−90
0
−10
phase
$|r_{xx}|$ dB
B
4 three objectives
B: in-phase bandwidth (%) ↑
L: lowest $|r_{xx}|$ in band (dB) ↑
D: fragile features ↓
band around 10 GHz
5 non-dominated archive (50 designs)
60
30
0
D
0
−10
−20
L (dB)
0
10
20
B (%)
archive
dataset front
dataset
6 improved PSO update
leader g: least-crowded cell of the archive
$\mathbf{v}_i \leftarrow w\mathbf{v}_i + c_1(\mathbf{p}_i - \mathbf{z}_i) + c_2(\mathbf{g} - \mathbf{z}_i)$, $\mathbf{z}_i \leftarrow \mathbf{z}_i + \mathbf{v}_i$
w: 0.9 → 0.4; $c_1 = 2.05|\cos 2\pi r|$, $c_2 = 2.05|\sin 2\pi r|$
mutation: half the swarm, step 10 % → 2 % of the range
next
generation
(T = 150–300)
final archive
(d) full-wave check of the selected designs
shortlist
after the final generation:
6 designs of the long VAE search
and 3 of the pixel search closest
to the ideal point
CST simulation
Floquet ports
2–12 GHz
t = 1.0 mm, ε_r = 2.65
prediction vs. CST
errors of $|r_{xx}|$ and phase;
B, L, D predicted and simulated
(Table VII, Fig. 8)